11 Skaters of masses 80 kg and 40 kg move directly towards each other and collide.

Before the collision, the heavier skater is moving to the right at a speed of $2.0\,m\,s^{-1}$ and the lighter skater is moving to the left at a speed of $1.0\,m\,s^{-1}$.

After the collision, the heavier skater moves to the right at a speed of $0.80\,m\,s^{-1}$.

What is the relative speed of separation of the two skaters?

A $0.6\,m\,s^{-1}$ **B** $1.4\,m\,s^{-1}$ **C** $2.2\,m\,s^{-1}$ **D** $2.6\,m\,s^{-1}$